



# AKTU

# B.Tech II-Year



# Cyber security



## Unit-4 in One Shot

## Pdf Notes + PYQs + Imp. Questions

## AKTU QUESTIONS

|     |                                                                                                        |                    |
|-----|--------------------------------------------------------------------------------------------------------|--------------------|
| 1.  | Explain the importance of chain of custody in digital forensic investigation                           | AKTU 2024-25       |
| 2.  | Describe the different method used for network forensic                                                | AKTU 2024-25       |
| 3.  | Explain the different stages of digital forensic life cycle                                            | AKTU 2024-25/23-24 |
| 4.  | What is chain of custody in forensic                                                                   | AKTU 2024-25       |
| 5.  | What are privacy threats? What are the challenge faced                                                 | AKTU 2023-24       |
| 6.  | What is Email? Explain how email forensic can be done                                                  | AKTU 2023-24       |
| 7.  | What is digital evidence                                                                               | AKTU 2023-24       |
| 8.  | What are the technical challenges by computer forensics professionals                                  | AKTU 2023-24       |
| 9.  | How does investigator capture and analysis the network traffic during network forensic investigations? | AKTU 2023-24       |
| 10. | What is digital forensic how it differ from tradition forensic science                                 | AKTU 2023-24       |

# What is Digital Forensics?

Digital Forensics is a branch of forensic science that deals with the **identification, preservation, analysis, and presentation** of evidence found on **digital devices** like computers, mobile phones, hard drives, USBs, cloud storage, and networks.

## Role of digital forensic

| Role                       | Description                                                                |
|----------------------------|----------------------------------------------------------------------------|
| 1.Crime Investigation      | Helps investigate cybercrimes like hacking, identity theft, cyberstalking. |
| 2.Data Recovery            | Recovers lost, deleted, or encrypted files during criminal investigations. |
| 3.Evidence Preservation    | Ensures data is not tampered with and is valid in court.                   |
| 4.Incident Response        | Identifies source and method of attack in a security breach.               |
| 5.Legal Support            | Provides expert testimony and digital evidence in court proceedings.       |
| 6.Corporate Investigations | Detects employee misconduct, policy violations, and insider threats.       |

| Aspect             | Traditional Forensics                                    | Digital Forensics                                   |
|--------------------|----------------------------------------------------------|-----------------------------------------------------|
| Definition         | Investigation of physical crimes using physical evidence | Investigation of cybercrimes using digital evidence |
| Evidence Type      | Blood, fingerprints, weapons, documents                  | Emails, files, logs, hard drives, mobile data       |
| Crime Scene        | Physical locations like homes, offices, crime spots      | Digital environments like computers, networks       |
| Tools Used         | Microscopes, fingerprint kits, DNA analyzers             | Forensic software like EnCase, FTK, Autopsy         |
| Data Nature        | Tangible and visible                                     | Intangible, stored in binary format                 |
| Investigation Time | Often slower due to lab testing                          | Faster if tools and data access are available       |
| Expertise Needed   | Biology, chemistry, physical evidence handling           | Computer science, cybersecurity, data recovery      |
| Tampering Evidence | Harder to alter physical evidence                        | Easier to alter or delete digital evidence          |
| Chain of Custody   | Documented physical transfer                             | Involves hashing and secure digital storage         |

## Example Activities digital forensic

- Imaging a suspect's hard disk without changing the original data.
- Analyzing browser history to trace illegal activity.
- Recovering deleted emails or messages.
- Tracing IP addresses in cybercrime.

## What is Computer Forensics?

Computer Forensics is a branch of digital forensics that focuses specifically on **investigating computers and digital storage devices** to find evidence of criminal or unauthorized activity.

## Things to Avoid During a Computer Forensic Investigation

| What to Avoid                       | Why                                                                     |
|-------------------------------------|-------------------------------------------------------------------------|
| 1. Powering on the system           | It may alter or destroy volatile evidence like RAM data.                |
| 2. Accessing files directly         | Opens risk of modifying timestamps or data accidentally.                |
| 3. Using original data for analysis | Can corrupt or overwrite evidence. Always use a forensic copy.          |
| 4. Not documenting actions          | Weakens the case in court; chain of custody breaks.                     |
| 5. Using non-forensic tools         | May alter metadata or fail to recover deleted/hidden data.              |
| 6. Ignoring volatile memory (RAM)   | Important live data like passwords or processes can be lost if ignored. |
| 7. Lack of legal permission         | Accessing or analyzing systems without proper authority is illegal.     |

# Things That Cannot Be Avoided

| What Cannot Be Avoided          | Why                                                                |
|---------------------------------|--------------------------------------------------------------------|
| 1. Proper Documentation         | Ensures evidence is credible and admissible in court.              |
| 2. Creating a forensic image    | Needed to preserve original data safely for analysis.              |
| 3. Maintaining chain of custody | Protects the integrity and authenticity of the evidence.           |
| 4. Use of forensic tools        | Specialized tools are required to extract and analyze data safely. |
| 5. Following legal procedures   | Ensures investigation is lawful and ethically sound.               |

# Technical Challenges Faced by Computer Forensics Professionals

| Challenge                        | Description                                                                     |
|----------------------------------|---------------------------------------------------------------------------------|
| 1. Data Encryption               | Encrypted files or drives are hard to access without passwords or keys.         |
| 2. Huge Volume of Data           | Modern devices store terabytes of data, making analysis time-consuming.         |
| 3. Deleted or Hidden Data        | Data may be intentionally deleted or hidden using steganography or rootkits.    |
| 4. Use of Anti-Forensic Tools    | Attackers use tools to erase, corrupt, or mislead forensic traces.              |
| 5. Cloud Storage                 | Evidence may be stored remotely, outside legal or physical reach.               |
| 6. Volatile Memory Analysis      | RAM data (passwords, sessions) disappears if not captured before shutdown.      |
| 7. Variety of Devices and OS     | Need to handle Windows, Linux, Android, iOS, IoT devices, etc.                  |
| 8. Frequent Technology Changes   | Constant updates in software and hardware require ongoing learning.             |
| 9. Legal and Jurisdiction Issues | Cloud data or emails may be hosted in another country, creating legal barriers. |

# Legal Challenges Associated with Computer Forensics

Computer forensic investigations must follow legal rules to ensure the evidence is valid and admissible in court

| Legal Challenge                     | Description                                                                              |
|-------------------------------------|------------------------------------------------------------------------------------------|
| 1. Privacy Laws                     | Accessing personal or corporate data may violate privacy rights or data laws.            |
| 2. Jurisdiction Issues              | Cybercrimes often cross borders, laws differ between countries or states.                |
| 3. Improper Search & Seizure        | Evidence collected without proper warrants can be ruled inadmissible.                    |
| 4. Chain of Custody Breaks          | If the handling of evidence isn't properly documented, it can be questioned.             |
| 5. Admissibility in Court           | Digital evidence must meet legal standards (authenticity, relevance, integrity).         |
| 6. Lack of Legal Authorization      | Investigating without consent or a court order may result in legal penalties.            |
| 7. Data Retention Laws              | Organizations may be required to keep or delete data , forensics must comply.            |
| 8. Cloud and Third-party Storage    | Accessing data on cloud servers owned by others may involve complex legal steps.         |
| 9. Use of Anti-Forensics by Accused | Suspects may delete or alter data, raising legal questions about intent and obstruction. |

## PHASES INVOLVED IN COMPUTER FORENSIC

| Phase             | Description                                                                     |
|-------------------|---------------------------------------------------------------------------------|
| 1. Identification | Locate potential sources of digital evidence (computers, phones, drives, etc.)  |
| 2. Preservation   | Secure and isolate the device to prevent data loss or tampering.                |
| 3. Collection     | Acquire digital evidence using forensic tools (e.g., disk imaging).             |
| 4. Examination    | Analyze data to find hidden, deleted, or suspicious files.                      |
| 5. Analysis       | Interpret the evidence to understand what happened, when, and who was involved. |
| 6. Documentation  | Record all steps taken and findings (needed for legal presentation).            |
| 7. Presentation   | Present the evidence and expert opinion in court or investigation reports.      |

## Why Computer Forensics is needed

| Purpose                 |                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------|
| Crime Investigation     | To detect and trace criminal activities like hacking, fraud, or cyberbullying.         |
| Data Recovery           | To retrieve deleted, lost, or encrypted files during legal or security investigations. |
| Legal Evidence          | To collect and present digital evidence in court in a valid and acceptable format.     |
| Cybersecurity           | To analyze how systems were compromised and prevent future attacks.                    |
| Internal Investigations | To identify employee misuse, data theft, or policy violations within organizations.    |
| National Security       | To detect terrorist activities or digital espionage through computer systems.          |

## What is Chain of Custody?

Chain of Custody is a **documented process** that tracks the **collection, handling, transfer, storage, and preservation** of evidence ,especially **digital evidence** from the moment it is found until it is presented in court.

### Key Components:

- Who collected the evidence
- When and where it was collected
- How it was stored or transferred
- Who accessed or analyzed it

## Importance of chain custody

| Reason                        | Explanation                                                                    |
|-------------------------------|--------------------------------------------------------------------------------|
| 1. Legal Validity             | Ensures the evidence is accepted in court by proving it was not tampered with. |
| 2. Evidence Integrity         | Maintains the originality and reliability of digital evidence.                 |
| 3.Accountability              | Tracks everyone who handled the evidence<br>no one can deny their involvement. |
| Transparency in Investigation | Builds trust in the investigation process and its findings.                    |

## What is Digital Evidence?

**Digital Evidence** is any information or data **stored or transmitted in digital form** that can be used in a court of law.

### Examples of Digital Evidence:

- Emails or chat messages
- Browser history
- Social media posts
- CCTV footage
- Files or images on a hard drive
- Call logs or SMS from a mobile phone

| Feature         | Digital Evidence                               | Physical Evidence                           |
|-----------------|------------------------------------------------|---------------------------------------------|
| Form            | Electronic data (files, logs, emails)          | Tangible items (weapons, fingerprints)      |
| Storage         | Stored in computers, mobile phones, USB drives | Found at crime scene (floor, clothes, etc.) |
| Handling        | Needs special tools/software to access safely  | Can be seen and handled physically          |
| Easily Changed? | Yes, can be edited or deleted                  | Harder to change without detection          |
| Copied Easily?  | Yes, can be copied exactly                     | No, copies may differ                       |
| Preservation    | Needs chain of custody and digital tools       | Needs secure packing and storage            |
| Analysis Tools  | Forensic software ( EnCase, FTK)               | Lab tools ( fingerprint kits, DNA test)     |

| S.No. | Best Practice for handling digital evidence | Explanation                                                          |
|-------|---------------------------------------------|----------------------------------------------------------------------|
| 1     | <b>Maintain Chain of Custody</b>            | Keep full record of who handled the evidence, when, and where.       |
| 2     | <b>Do Not Alter Original Data</b>           | Never open or change the original file/device work on a copy only.   |
| 3     | <b>Use Forensic Tools</b>                   | Use trusted tools (FTK, EnCase) to avoid damaging the data.          |
| 4     | <b>Protect from Damage</b>                  | Store safely using anti-static bags; keep away from heat or water.   |
| 5     | <b>Avoid Booting Suspect Device</b>         | Bootting can change data use write-blockers to prevent changes.      |
| 6     | <b>Document Everything</b>                  | Write down every step, take photos, and record what you find and do. |
| 7     | <b>Restrict Access</b>                      | Only trained experts should handle the evidence.                     |
| 8     | <b>Follow Legal Rules</b>                   | Always get proper permission (like a warrant) to collect evidence.   |

## What is the Rule of Evidence?

The **Rule of Evidence** refers to **legal principles** that govern how evidence is **collected, presented, and accepted** in a court of law.

To ensure that only **relevant, reliable, and lawful** evidence is used in legal proceedings.

| Key Rules of Evidence | Explanation                                                                         |
|-----------------------|-------------------------------------------------------------------------------------|
| Relevance             | Evidence must be related to the case.                                               |
| Admissibility         | Evidence must be legally acceptable (not illegally obtained).                       |
| Authenticity          | Evidence must be genuine and verifiable.                                            |
| Best Evidence Rule    | Original documents are preferred over copies.                                       |
| Hearsay Rule          | Second-hand statements are usually not allowed unless exceptions apply.             |
| Chain of Custody      | Proper documentation of who handled the evidence to ensure it wasn't tampered with. |

## What is Network Forensics?

Network forensics is the process of **capturing, recording, and analyzing network traffic** to detect and investigate **unauthorized access, cyberattacks, or data breach**

| Method used for capturing network traffic | Description                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Packet Sniffing</b>                    | Tools like <b>Wireshark</b> or <b>tcpdump</b> capture real-time data packets on a network. |
| <b>Network Taps or Port Mirroring</b>     | Devices or switch configurations are used to <b>mirror traffic</b> to forensic tools.      |
| <b>Log Collection</b>                     | Collect logs from <b>firewalls, routers, IDS/IPS, or servers</b> for investigation.        |
| <b>Flow Data (NetFlow/IPFIX)</b>          | Captures metadata like IPs, ports, duration — not full packet content.                     |
| <b>Cloud Traffic Monitoring</b>           | Use cloud-native tools ( AWS VPC Flow Logs) for cloud-based forensics.                     |

## Method for network traffic analysis

### Description

#### Protocol Analysis

Identify suspicious protocols or abnormal usage ( HTTP over unusual ports).

#### Content Inspection

Examine packet contents for malware, commands, or data exfiltration.

#### Traffic Pattern Analysis

Detect spikes, unusual timings, or large file transfers.

#### Correlation with Logs

Match captured data with logs ( login attempts, alerts) to reconstruct events.

#### Timeline Reconstruction

Rebuild sequence of events during the attack using timestamps.

## Important Elements in a Forensic Investigation Engagement Contract

An engagement contract outlines the **scope, responsibilities, and legal terms** between a forensic investigator (or firm) and the client. Key elements include

| Element                        | Description                                                            |
|--------------------------------|------------------------------------------------------------------------|
| 1. Scope of Work               | Defines what will be investigated (data breach, fraud, digital theft). |
| 2. Objectives                  | What the client wants to achieve (evidence collection, root cause).    |
| 3. Legal Authorization         | Confirms the investigator has permission to access systems and data.   |
| 4. Confidentiality Clause      | Ensures all sensitive information remains private and protected.       |
| 5. Chain of Custody Procedures | Describes how evidence will be handled, stored, and documented.        |
| 6. Tools and Methods Used      | Lists forensic tools/methodologies to be applied in the investigation. |

# Important Elements in a Forensic Investigation Engagement Contract

| Element                       | Description                                                                |
|-------------------------------|----------------------------------------------------------------------------|
| 7. Roles and Responsibilities | Defines who will do what ( client vs. investigator.)                       |
| 8. Deliverables               | Specifies reports, evidence logs, or expert opinions to be provided.       |
| 9. Timeframe                  | Sets deadlines for investigation phases and final report submission.       |
| 10. Fees and Payment Terms    | Agreed costs, billing methods, and payment schedule.                       |
| 11. Liability Limitations     | Defines the extent of responsibility in case of data loss or legal issues. |
| 12. Dispute Resolution        | Outlines how conflicts will be resolved                                    |
| 13. Termination Clause        | Conditions under which the contract can be ended early.                    |

# What are Privacy Threats?

**Privacy threats** are risks that compromise a person's or organization's private information. These threats can expose personal data, such as name, address, passwords, health records, financial details, etc., without permission.

| Type of Privacy Threat  | Explanation                          | Example                                     |
|-------------------------|--------------------------------------|---------------------------------------------|
| Phishing                | Fake messages to steal personal info | Fake email asking for bank details          |
| Data Breaches           | Unauthorized access to data          | Hackers stealing user info from a company   |
| Spyware                 | Software that secretly collects data | Apps tracking your activity without consent |
| Social Engineering      | Manipulating people to give up info  | Impersonating IT staff to get passwords     |
| Tracking & Profiling    | Online behavior is monitored         | Websites tracking browsing habits           |
| Cloud Misconfigurations | Improper cloud settings expose data  | Public cloud files without access control   |

## What is a Social Networking Site?

A **social networking site** is an online platform that allows people to create profiles, connect with others, share content, and communicate.

**Examples** Facebook, Instagram, Twitter ,LinkedIn, Snapchat

## Security Threats from Social Networking Sites

| Threat             | Explanation                                             | Example                                                                      |
|--------------------|---------------------------------------------------------|------------------------------------------------------------------------------|
| Identity Theft     | Criminals gather your personal info to impersonate you. | Using your photos and details to create fake accounts.                       |
| Phishing Attacks   | Fake links or messages trick users into giving up data. | A message saying “click here to win an iPhone” leading to a fake login page. |
| Malware Spread     | Harmful links or apps are shared through posts or DMs.  | Clicking on a video link that downloads spyware.                             |
| Social Engineering | Attackers use info from your profile to manipulate you. | “Hi, I’m your cousin, can you send me money?”                                |

# Security Threats from Social Networking Sites

| Threat                | Explanation                                             | Example                                                         |
|-----------------------|---------------------------------------------------------|-----------------------------------------------------------------|
| Stalking & Harassment | Strangers misuse personal details.                      | Posting your location publicly and someone shows up there.      |
| Fake Profiles         | Scammers create false accounts to scam or trick others. | Fake job recruiter asking for bank details.                     |
| Oversharing           | Users reveal too much info unintentionally.             | Posting birthdate, school, and pet name—all hints to passwords. |

## What is Email?

Email (Electronic Mail) is a method of exchanging digital messages over the internet between people using electronic devices (like computers, phones). It uses standard protocols like SMTP (sending), POP3 or IMAP (receiving).

## What is Email Forensics?

Email Forensics is the process of investigating email messages to collect digital evidence, mainly for cybercrime investigations (like phishing, fraud, threats, etc.).

## How Email Forensics is Done

| Step                      | Explanation                                                                     |
|---------------------------|---------------------------------------------------------------------------------|
| 1. Collection             | Securely collect the email from server, client, or backup without tampering.    |
| 2. Header Analysis        | Examine the email header (metadata) to trace source IP, route, and sender info. |
| 3. Body Analysis          | Analyze message content for threats, malicious links, or spam.                  |
| 4. Attachment Analysis    | Scan attachments for malware, hidden scripts, or trojans.                       |
| 5. Keyword Search         | Use tools to find suspicious keywords or phrases in emails.                     |
| 6. Timestamp Verification | Check email date/time stamps to detect tampering.                               |
| 7. Metadata Analysis      | Extract hidden info (software used, file history).                              |
| 8. Trace Email Path       | Reconstruct the path ,the email took across networks.                           |
| 9. Reporting              | Document findings in a legal report for court or investigation.                 |

## What is an Email Message Header?

An **email header** is the part of the email that contains **technical details** about the email's journey. It is **not usually visible** to the user but is used in **email forensics** to trace the source and path of the email.

| Field                  | Meaning                                                               |
|------------------------|-----------------------------------------------------------------------|
| <b>From:</b>           | Email address of the sender                                           |
| <b>To:</b>             | Email address of the receiver                                         |
| <b>Subject:</b>        | The topic/title of the email                                          |
| <b>Date:</b>           | When the email was sent                                               |
| <b>Return-Path:</b>    | Address to which undeliverable emails should go                       |
| <b>Received:</b>       | Shows each server the email passed through (used to trace email path) |
| <b>Message-ID:</b>     | A unique ID given to the email by the sender's system                 |
| <b>Content-Type:</b>   | Format of the email content (text, HTML, attachment, etc.)            |
| <b>MIME-Version:</b>   | Version of the MIME protocol (for handling attachments and formats)   |
| <b>SPF/DKIM/DMARC:</b> | Email authentication results (used to detect spoofing or spam)        |

# Explain the different stages of digital forensic life cycle

| Stage             | Description                                                                                                                                                                             | Detailed Example                                                                                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identification | <b>Detection of the Incident:</b> This stage is about recognizing that a security breach or digital crime has occurred. It involves monitoring systems and detecting anomalies.         | An IT team notices unusual outbound network traffic and multiple failed login attempts on a company server. They suspect unauthorized access is taking place.                                          |
| 2. Preservation   | <b>Securing the Evidence:</b> Once an incident is identified, the priority is to secure the digital evidence in its original state to prevent any tampering or accidental modification. | The forensic team isolates the affected server from the network to stop further activity. They then create a bit-for-bit image (exact copy) of the hard drive to work on, keeping the original intact. |
| 3. Collection     | <b>Gathering Digital Data:</b> In this phase, all relevant digital data from multiple sources (devices, logs, emails, etc.) is systematically collected.                                | The team collects data from the compromised server, including system logs, application logs, and network logs etc                                                                                      |

# Explain the different stages of digital forensic life cycle

| Stage            | Description                                                                                                                                                                                                   | Detailed Example                                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. Examination   | <b>Detailed Analysis of the Data:</b> Specialized forensic tools are used to sift through the collected data. This includes recovering deleted files, analyzing file metadata, and checking system artifacts. | Using forensic software like Autopsy or FTK, investigators analyze the disk image of the server to recover deleted files, review login histories, and inspect file timestamps to establish a timeline.                       |
| 5. Analysis      | <b>Interpreting the Evidence:</b> The data is interpreted and correlated to understand the nature, scope, and methodology of the incident. Patterns, anomalies, and timelines are built here.                 | Analysts correlate recovered files, network logs, and user activity. They determine that the intruder exploited a software vulnerability to gain access, then used stolen credentials to move laterally within the network.  |
| 6. Documentation | <b>Recording Findings:</b> Every step taken during the investigation is meticulously documented to maintain a clear chain of custody. This documentation is vital for legal and reporting purposes.           | Investigators keep a detailed log describing when and how the evidence was collected, the tools used, the steps taken during the examination, and notes on all findings to ensure transparency in the investigation process. |

# Explain the different stages of digital forensic life cycle

| Stage           | Description                                                                                                                                                                                                   | Detailed Example                                                                                                                                                                                                                                            |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7. Presentation | <b>Reporting the Outcome:</b> The results of the investigation are compiled into a comprehensive report. The findings are presented in a format suitable for technical teams, management, and legal entities. | A final report is drafted that outlines the timeline of the attack, the methods used by the attacker, all evidence collected, and recommendations for improving security. This report may be presented in court or to management to support further action. |
| 8. Review       | <b>Post-Investigation Reflection:</b> The process is reviewed to identify strengths and weaknesses in the investigation. Lessons learned are used to refine future security and forensic procedures.          | After the case is closed, a debrief is held. The team reviews how quickly and accurately the threat was identified, preserved, and analyzed, suggesting improvements such as enhanced monitoring tools and updated incident response protocols.             |

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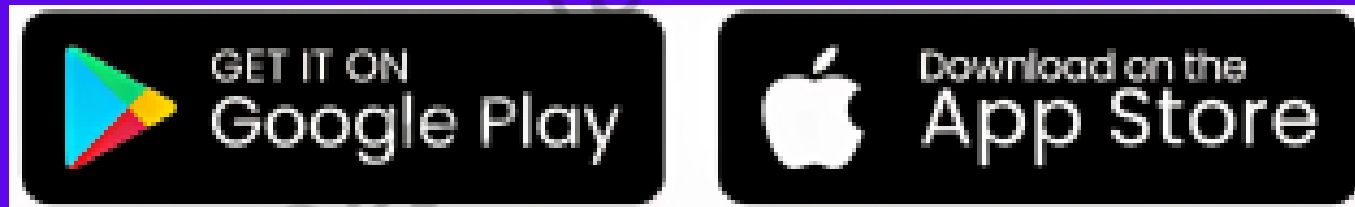


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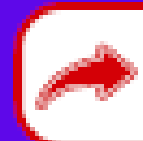


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